Linear Maps: Free Variables, Obstructions, and Canonical Splittings

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A matrix equation $Ax = b$ raises two logically independent questions. Does a solution exist? If it does, is it unique? The image answers the first question, and the kernel answers the second. This elementary observation leads to exact sequences and quotient spaces; with an inner product, it leads further to the pseudoinverse and an optimal choice among the many possible solutions.[1]

§1.1 Free variables describe the kernel as a graph

Consider the row-reduced matrix

$$R = \begin{pmatrix} 1 & 3 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 1 & -4 \end{pmatrix}.$$

Its pivot columns are $1, 3, 4$; columns $2, 5$ are free. The system $Rx = 0$ reads

$$x_1 = -3x_2 - 3x_5, \qquad x_3 = -9x_5, \qquad x_4 = 4x_5.$$

Thus

$$\boxed{\ker R = \operatorname{span}\left\{ \begin{pmatrix} 3 \\ -1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ 9 \\ -4 \\ -1 \end{pmatrix} \right\}.}$$

The negative signs in the free coordinates are immaterial; they merely give the negatives of the usual kernel basis vectors. This is the practical minus-one trick.

The structural calculation is more useful than the trick. After permuting pivot coordinates before free coordinates, every RREF system has the form

$$\begin{pmatrix} I_r & C \end{pmatrix} \begin{pmatrix} x_p \\ x_f \end{pmatrix} = 0.$$

Therefore

$$\boxed{\ker \begin{pmatrix} I_r & C \end{pmatrix} = \left\{ \begin{pmatrix} -Cu \\ u \end{pmatrix} : u \in \mathbb{F}^{n-r} \right\}.}$$

The kernel is the graph of the map $u \mapsto -Cu$. Its dimension is $n - r$ because the free coordinate u can be chosen arbitrarily.

A matrix changes with the bases in its domain and codomain. If the new coordinate matrix is $T^{-1}RS$, with S, T invertible, its kernel is $S^{-1} \ker R$. Consequently kernel *dimension* and rank are basis-independent, even though the displayed basis vectors change.

[1] For the linear-map, quotient, orthogonal-complement and pseudoinverse framework see Sheldon Axler, *Linear Algebra Done Right*, Chapters 3 and 6.

§1.2 Rank–nullity is the dimension of an exact sequence

For any linear map $T : V \to W$, factor it as

$$V \xrightarrow{q} V/\ker T \xrightarrow{\overline{T}} \operatorname{im} T \hookrightarrow W,$$

where q sends v to its equivalence class and

$$\overline{T}([v]) = T(v).$$

This is well defined: if $v - v' \in \ker T$, then $T(v) = T(v')$. It is also injective and surjective onto the image, so

$$\boxed{V/\ker T \cong \operatorname{im} T.}$$

In finite dimensions, counting dimensions gives

$$\boxed{\dim V = \dim \ker T + \dim \operatorname{im} T.}$$

The underlying exact sequence is

$$0 \longrightarrow \ker T \longrightarrow V \xrightarrow{T} \operatorname{im} T \longrightarrow 0.$$

Exactness means that the image of each arrow is precisely the kernel of the next. It is not a decorative way of displaying rank–nullity: it remains meaningful in settings where numerical dimensions are unavailable.

§1.3 The cokernel measures equations that cannot be solved

The cokernel of $T : V \to W$ is

$$\operatorname{coker} T = W/\operatorname{im} T.$$

A right-hand side $b \in W$ is in the image of T precisely when its class in the cokernel is zero.

To see this computationally, consider

$$B = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}.$$

Every output of B lies on the line

$$(u, v) = (t, 2t), \qquad t \in \mathbb{R}.$$

The vector $y = (-2, 1)^T$ spans the left null space because $y^T B = 0$. Hence the equation $Bx = b$ is solvable exactly when

$$\boxed{-2b_1 + b_2 = 0.}$$

This condition is both necessary and sufficient: if $b_2 = 2b_1$, simply take $x = (b_1, 0, 0)^T$.

Here

$$\dim \ker B = 2, \qquad \dim \operatorname{im} B = 1, \qquad \dim \operatorname{coker} B = 1.$$

The two-dimensional kernel records the nonuniqueness of solutions, while the one-dimensional cokernel records the single compatibility condition on b.

In finite-dimensional inner-product spaces,

$$\boxed{(\operatorname{im} T)^{\perp} = \ker T^{*}.}$$

Indeed, $w \perp \operatorname{im} T$ exactly when $\langle Tv, w\rangle = 0$ for all v, or $T^{*}w = 0$. This gives a concrete realization of cokernel constraints by left null vectors.

§1.4 An arbitrary solution is not an optimal solution

A surjective linear map admits many right inverses, because one may choose many complementary subspaces to its kernel. An inner product singles out one canonical choice: the solution orthogonal to the kernel.

Take

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Its kernel is spanned by $k = (1, -1, 1)^T$, since $Ak = 0$. The matrix has full row rank, so AA^T is invertible:

$$AA^T = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad (AA^T)^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

The Moore–Penrose right inverse is

$$\boxed{A^+ = A^T(AA^T)^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ 1 & 1 \\ -1 & 2 \end{pmatrix}.}$$

Direct multiplication gives $AA^+ = I_2$.

For $b = (1, 2)^T$, the preferred solution is

$$x_* = A^+ b = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

All solutions are

$$x = x_* + tk, \qquad t \in \mathbb{R}.$$

Since $x_* \cdot k = 0$, Pythagoras gives

$$\|x\|^2 = \|x_*\|^2 + t^2\|k\|^2 = 2 + 3t^2.$$

Thus x_* is the unique solution of minimum Euclidean norm. Choosing the free coordinate $x_2 = 0$ would instead produce $(1, 0, 2)^T$, whose squared norm is 5, strictly larger than 2.

The orthogonal splitting can also be read from two projectors:

$$P = A^+A, \qquad Q = I_3 - P.$$

They satisfy

$$P^2 = P = P^T, \qquad Q^2 = Q = Q^T, \qquad PQ = 0.$$

Here P projects onto the row space $\operatorname{im} A^T$, and Q projects onto $\ker A$. Row reduction selects one arbitrary complement of the kernel; the Euclidean inner product singles out its orthogonal complement.

§1.5 A dimension identity survives the choice of rank

Combine the kernel and cokernel in the longer exact sequence

$$0 \longrightarrow \ker T \longrightarrow V \xrightarrow{T} W \longrightarrow \operatorname{coker} T \longrightarrow 0.$$

Taking the alternating sum of dimensions gives

$$\boxed{\dim \ker T - \dim \operatorname{coker} T = \dim V - \dim W.}$$

For either of the 2×3 matrices above the right side equals 1. The surjective matrix A has kernel dimension 1 and cokernel dimension 0, whereas the rank-one matrix B has kernel dimension 2 and cokernel dimension 1. Both yield the same difference 1.

This finite-dimensional invariant motivates the *Fredholm index* of suitable linear operators between infinite-dimensional spaces. There one requires finite-dimensional kernel and cokernel and closed range, so their difference remains meaningful even though domain and codomain may both have infinite dimension. The definition records the same balance between nonuniqueness and obstruction that began in a row-reduction exercise.

The computation has moved through several compatible descriptions. Free variables parametrize the kernel; the first isomorphism theorem identifies the information that survives the map; left null vectors detect unreachable right-hand sides; and the pseudoinverse uses an inner product to choose the unique minimum-norm solution. Exact sequences express what all these calculations have in common without requiring a preferred coordinate system.